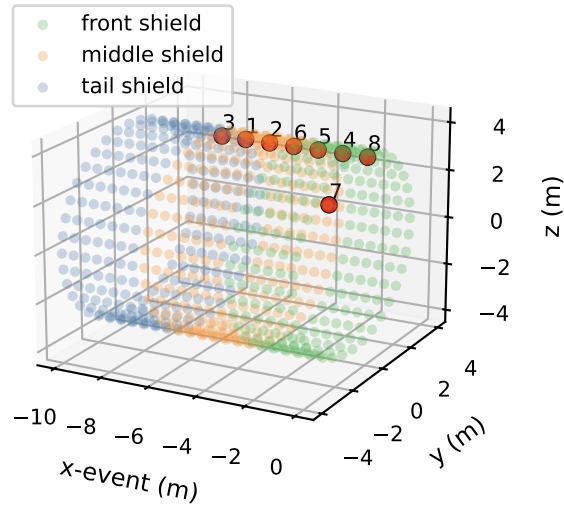
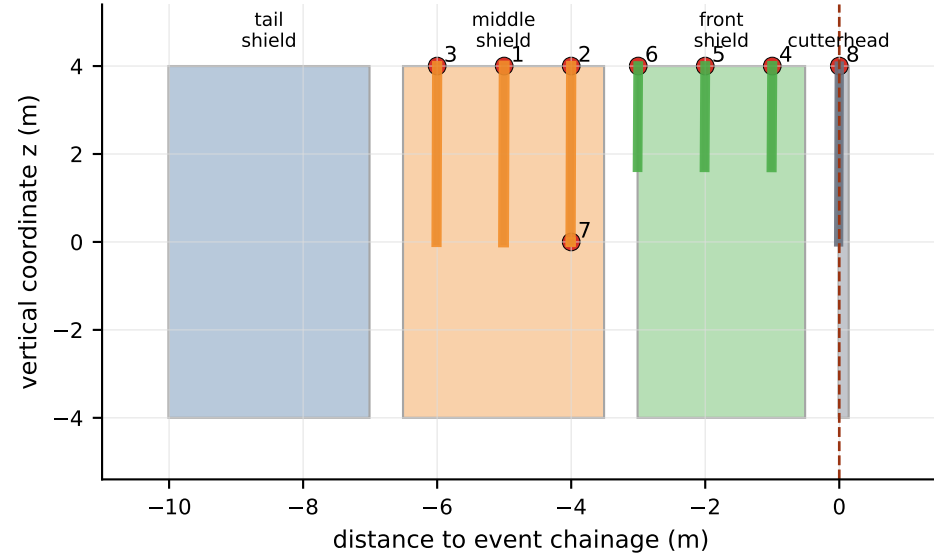


A. Local 3D rock-shield edge provenance



B. Longitudinal section of top contributing rock-shield edges



C. Top edge-level provenance records near the stuck state

Rank	Component	Rel. chainage	Distance	Normal	q_i	w_ij(t)	Contribution
1	middle shield	-5.000	0.051	0.982	0.891	0.958	0.853
2	middle shield	-4.000	0.051	0.982	0.829	0.958	0.794
3	middle shield	-6.000	0.051	0.982	0.820	0.958	0.785
4	front shield	-1.000	0.051	0.982	0.806	0.958	0.772
5	front shield	-2.000	0.051	0.982	0.783	0.958	0.750
6	front shield	-3.000	0.051	0.982	0.768	0.958	0.735
7	middle shield	-4.000	0.051	0.982	0.734	0.958	0.703
8	cutterhead	0.000	0.010	1.000	0.694	0.995	0.691